Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 4: If, Then

The Implication

Chartrand & Zhang, Section 1.3

Name: _____ Date: _____

Where we are

On Friday you simplified $\sim\big(P\wedge(\sim Q)\big)$ and got $(\sim P)\vee Q$ — a statement meaning *"it never happens that P holds while Q fails."* You were told to keep that page. Today that compound statement gets its name.

By the end of today you should be able to

1. State the truth table for $P \Rightarrow Q$ and say when it is false.
2. Explain why an implication with a false hypothesis is true.
3. Recognise a vacuously true statement.
4. Negate an implication correctly.

1. The definition

Definition 1.33 — implication

For statements P and Q, the **implication** $P \Rightarrow Q$ is the statement

$$\text{If } P, \text{ then } Q.$$

P is the **hypothesis**; Q is the **conclusion**. An implication is also called a *conditional*.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

$P \Rightarrow Q$ is false in exactly one row: when the hypothesis is true and the conclusion is false.

Compare with Friday

Put this column next to the one you built for $(\sim P) \vee Q$. They are identical.

$$P \Rightarrow Q \;\equiv\; (\sim P) \vee Q \qquad \text{(Theorem 1.48)}$$

The implication is not a new idea bolted onto logic. It is a combination of $\sim$ and $\vee$ that turns up so often it earned its own symbol.

2. Why a false hypothesis makes it true

This is the row that bothers everyone. Think of $P \Rightarrow Q$ as a **contract**.

A neighbour says: *"If you pay me an extra \$50, then I will also paint your dining room."*

When has the neighbour **broken** the promise?

You pay \$50?	Dining room painted?	Promise broken?
Yes	Yes	No — kept
Yes	No	**Yes — broken**
No	Yes	No — painted anyway, no promise violated
No	No	No — the condition never arose

Only one row breaks the promise, and it is the same row that makes the implication false. **A promise you were never called on to keep has not been broken.**

What the truth table does not claim

$P \Rightarrow Q$ being true does *not* mean P causes Q, or that the two are related at all. "If $2+2=5$, then the moon is made of cheese" is a **true** statement, because its hypothesis is false.
That feels wrong on first meeting. It is the price of having a definition that depends only on truth values — and outside contrived examples, hypotheses and conclusions are almost always genuinely connected.

3. Vacuous truth — a return visit

Practice Set 1 ended with this, and most of the class wanted to answer "neither":

> *Every student in this room who is over 200 years old has a Ph.D.*

Now you can settle it. For each student, the claim is *if over 200 years old, then has a Ph.D.* Since nobody in the room satisfies the hypothesis, every one of those implications sits in row 3 or 4 of the table, and is therefore true.

A claim about all members of an empty collection is called **vacuously true**.

This is why `all([])` returns `True`

```
all([]) # True -- nothing violates the condition
all(x > 100 for x in []) # True -- same reason
```

Every language with an "all" operation makes this choice, and it is the same convention for the same reason.

4. Negating an implication

If $P \Rightarrow Q$ is false in exactly one row, then its negation is true in exactly that one row. So the negation must say: *P holds **and** Q fails.*

Theorem 1.50

$$\sim(P \Rightarrow Q) \;\equiv\; P \wedge (\sim Q)$$

You can also derive it from what you already know:

Expression	Law used
$\sim(P \Rightarrow Q)$	(start)
$\equiv \sim((\sim P) \vee Q)$	Theorem 1.48
$\equiv (\sim(\sim P)) \wedge (\sim Q)$	De Morgan (Thm 1.24)
$\equiv P \wedge (\sim Q)$	Double negation (Thm 1.28)

The negation of an implication is not an implication

The negation of "if it rains, the game is cancelled" is **not** "if it rains, the game is not cancelled." It is **"it rains and the game is not cancelled"** — the single situation that would break the promise. Watch for this. It is the most common error in Section 1.3, and it is the whole content of Theorem 1.50.

Practise it. Write the negation of each.

Statement	Negation
If the input is sorted, the search succeeds.	The input is sorted and the search does not succeed.
If n is prime, then n is odd.	n is prime and n is not odd. (True: $n = 2$.)

Symbols so far

Symbol	Name	False only when
$\sim P$	negation	P is true
$P \wedge Q$	conjunction	at least one is false
$P \vee Q$	disjunction	both are false
$P \oplus Q$	exclusive or	both alike
$P \Rightarrow Q$	implication	P true and Q false

Exit ticket

1. Is "If $5 < 3$, then 7 is even" true or false? Explain in one line.

2. Write the negation of "If the file exists, then it can be read."

Before Wednesday

Read Section 1.3, pp. 28–31 — the ways of expressing an implication in words (Figure 1.10), the converse, and the contrapositive.

Self-check (not collected). Section 1.3, Exercises 1 and 3. These are odd-numbered, so answers are in the back of the book — attempt them fully before looking.

Homework 2 is due Friday, August 28, 11:59 p.m.

Questions I still have